

Personal AI Clone — Sprint 2 Report

Title Page

Project: Personal AI Clone

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Section: C **Dates:** Nov 01 – Nov 14, 2025

Tasks

- Decide fine-tuning approach depending on data size:
 - Plenty of text ($\geq 50k$ tokens): full fine-tuning (GPT-2 small / distil-GPT2) or LLaMA-like checkpoint
 - Limited data ($< 50k$ tokens): LoRA / adapter tuning + prompt engineering
 - Very small data: RAG + prompt templates
- Decide model provider: Local (Hugging Face + PyTorch) or hosted API (OpenAI)
- Create `docs/model_design.md` explaining chosen method + rationale

Tasks:

- Prepare backend and training environments
- Add required packages in:
 - `backend/requirements.txt`
 - `training/requirements.txt`
- Create virtual environment and install dependencies

Files created/updated:

- `backend/requirements.txt`
- `training/requirements.txt`
- `.venv/` folder (local virtual environment)

Tasks:

- Load only approved raw data (do NOT commit raw files)
- Normalize text and remove PII
- Split dialogs/messages into prompt→response pairs
- Create JSONL training examples
- Shuffle and split into train/validation sets
- Save processed data to:
 - `data/processed/user_corpus.jsonl`
 - `data/processed/user_corpus_val.jsonl`

Files created/updated:

- `scripts/preprocess.py`
- `notebooks/data_preprocessing.ipynb`
- `data/processed/user_corpus.jsonl`
- `data/processed/user_corpus_val.jsonl`

Tasks:

- Tokenize dataset using selected tokenizer
- Inspect dataset statistics:
 - Number of examples
 - Average tokens per example
 - Vocabulary issues
- Concatenate messages if average tokens per example is too small

Files created/updated:

- `notebooks/data_preprocessing.ipynb` (updated with tokenization and stats)

Tasks:

- Choose training method:
 - Recommended: LoRA + 8-bit training for small datasets
- Prepare small proof-of-concept training run
- Adjust hyperparameters for hardware constraints

Files created/updated:

- `training/train.py`

Tasks:

- Run training on a small dataset to validate pipeline
- Monitor logs for errors; reduce batch size if necessary
- Save logs and loss/validation loss plots:
 - `logs/train_run_YYYYMMDD.txt`
 - `docs/loss_plots.png` (or any chart format)
- Create reproducible `requirements.txt`

Files created/updated:

- `logs/train_run_YYYYMMDD.txt`
- `docs/` (plots and documentation)

Tasks:

- Document dataset size (#examples, #tokens)
- Record hyperparameters and training/validation loss
- Generate inference examples
- Compute metrics:
 - Perplexity
 - Embedding similarity (SBERT)
 - Human-rated evaluation (if possible)

Files created/updated:

- `docs/training_report.md`

Tasks:

- Save final model to `models/user_clone/`
 - If using LoRA, save base model and adapters
- Optionally upload to Hugging Face Hub (private repo)
- Ensure folder contains all necessary files:
 - `config.json`
 - `pytorch_model.bin`
 - Adapter files (if using PEFT/LoRA)

Files created/updated:

- `models/user_clone/` (all model files)

Tasks:

- Create FastAPI server to serve the model
- Load tokenizer and model
- Expose endpoint `/generate` for inference
- Test server with demo client
- Handle CPU/GPU device

Local URLs for reference:

- Swagger API docs: [FastAPI Docs](#)

Files created/updated:

- `backend/model_server.py`
- `demo/generate_sample.py`

Tasks:

- Create small UI (React or HTML) to send requests to `/generate`
- Handle prompt input and display generated output
- Configure proxy if frontend/backend run on same host

Frontend Demo URL (local server):

- [Demo UI](#)

Files created/updated:

- `frontend/src/App.jsx` (or `frontend/ui.html` if simple HTML)

Day K — Safety & Guardrails (Nov 11)

Tasks:

- Add output filtering pipeline (toxicity check)
- Implement human-in-the-loop for text approval
- Log prompts and outputs securely (do NOT commit logs to VCS)
- Integrate safety functions into inference API

Files created/updated:

- `backend/safety.py`
- Updates in `backend/model_server.py` to use safety filters